

FitHub AI

“Train smarter. Live better.”

Comprehensive Software Project Life Cycle Document

Project Name	FitHub AI
Application Type	AI-Powered Fitness Coaching Web Application
Primary Tech Stack	Python (FastAPI), Jinja2, SQLAlchemy, SQLite, Google Gemini API, Docker, Uvicorn

5. Project Development Phase

Directory Structure Implementation

```
fitbuddy-ai.nit/
|-- app/
|   |-- __init__.py
|   |-- config.py           # Dynamic environment configuration
|   |-- main.py             # FastAPI initialization & entry point
|   |-- routes.py           # Route definitions and handlers
|   |-- template/           # UI templates
|       |-- base.html
|       |-- home.html
|       |-- dashboard.html
|       |-- planner.html
|       |-- contact.html
|       |-- index.html
|-- .dockerignore
|-- Dockerfile
|-- Procfile
|-- render.yaml
|-- deploy_vps.sh           # VPS automation script
|-- start.ps1               # Local startup script
|-- requirements.txt        # Python dependencies
`-- runtime.txt
```

Key Development Tasks Completed

- Configured config.py to securely read runtime settings from .env.
- Built routes.py handlers to parse multi-variable HTTP form submissions from planner.html and invoke Gemini API outputs.
- Automated local startup using PowerShell (start.ps1) and created automated Linux server setup scripts (deploy_vps.sh).